

Worked Examples: Golden Section & Fibonacci Search

ISE 5406 — Optimization 2 | Lecture 10 Companion

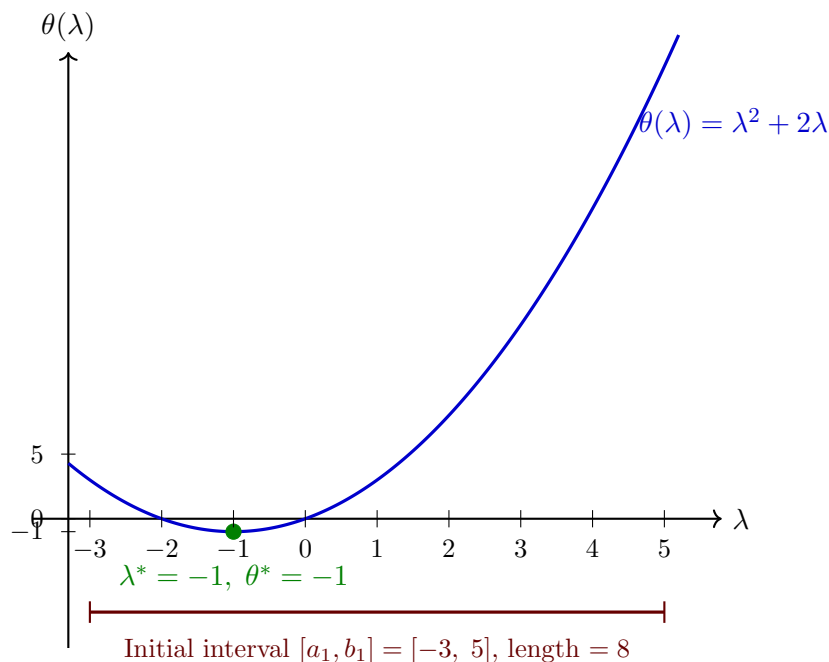
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Problem Setup

We wish to minimize the unimodal function

$$\theta(\lambda) = \lambda^2 + 2\lambda$$

on the interval $[a_1, b_1] = [-3, 5]$. The true minimum is at $\lambda^* = -1$ with $\theta(\lambda^*) = -1$.



1 Golden Section Search

Key Idea

The golden section method places two test points using the **golden ratio** $\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618$, which is the same at every iteration. After comparing function values, one side of the interval is eliminated and one test point is *reused*, so each iteration requires only **one new function evaluation**.

Placement formulas. At iteration k with current interval $[a_k, b_k]$:

$$\lambda_k = \alpha a_k + (1 - \alpha) b_k, \quad \mu_k = (1 - \alpha) a_k + \alpha b_k.$$

Iteration 1

Current interval: $[a_1, b_1] = [-3, 5]$, length = 8.

Step 1 — Place test points:

$$\lambda_1 = \alpha a_1 + (1 - \alpha) b_1 = 0.618(-3) + 0.382(5) = \mathbf{0.056}$$

$$\mu_1 = (1 - \alpha) a_1 + \alpha b_1 = 0.382(-3) + 0.618(5) = \mathbf{1.944}$$

Note that λ_1 and μ_1 are placed at equal distances from the endpoints: both are a fraction α of the interval length from one end.

Step 2 — Evaluate:

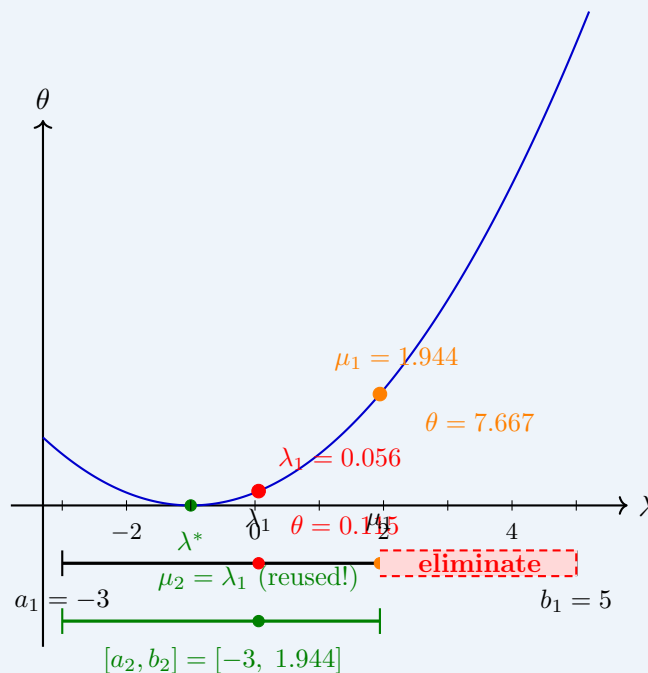
$$\theta(\lambda_1) = (0.056)^2 + 2(0.056) = \mathbf{0.115}$$

$$\theta(\mu_1) = (1.944)^2 + 2(1.944) = \mathbf{7.667}$$

Step 3 — Compare & eliminate: Since $\theta(\lambda_1) = 0.115 \leq 7.667 = \theta(\mu_1)$, the minimum cannot lie in $(\mu_1, b_1]$ (by unimodality). Discard the right portion:

$$[a_2, b_2] = [a_1, \mu_1] = [-3, 1.944], \quad \text{length} = 4.944.$$

Key efficiency: The point $\lambda_1 = 0.056$ automatically becomes μ_2 in the new interval, so $\theta(\mu_2) = 0.115$ is *already known* — no new function evaluation needed for μ_2 .



Iteration 2

Current interval: $[a_2, b_2] = [-3, 1.944]$, length = 4.944.

Known: $\mu_2 = \lambda_1 = 0.056$ with $\theta(\mu_2) = 0.115$ (reused).

Compute new point:

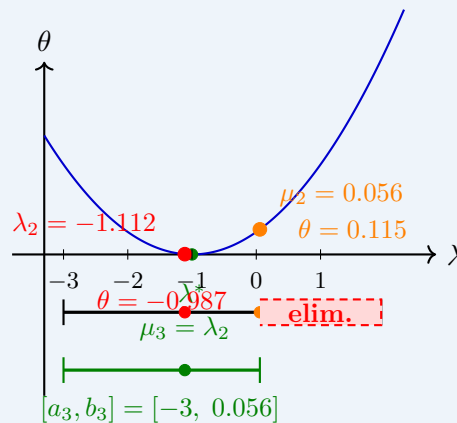
$$\lambda_2 = 0.618(-3) + 0.382(1.944) = -1.112$$

$$\theta(\lambda_2) = (-1.112)^2 + 2(-1.112) = 1.237 - 2.224 = -0.987$$

Compare: $\theta(\lambda_2) = -0.987 \leq 0.115 = \theta(\mu_2)$. Eliminate right.

$$[a_3, b_3] = [-3, 0.056], \text{ length} = 3.056.$$

Reuse: $\mu_3 = \lambda_2 = -1.112$, $\theta(\mu_3) = -0.987$.



Iteration 3

Current interval: $[a_3, b_3] = [-3, 0.056]$, length = 3.056.

Known: $\mu_3 = \lambda_2 = -1.112$ with $\theta(\mu_3) = -0.987$ (reused).

Compute new point:

$$\lambda_3 = 0.618(-3) + 0.382(0.056) = -1.832$$

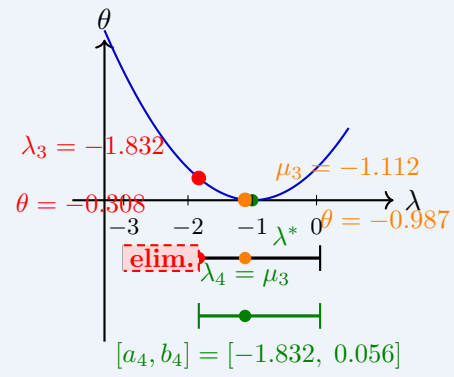
$$\theta(\lambda_3) = (-1.832)^2 + 2(-1.832) = 3.356 - 3.664 = -0.308$$

Compare: $\theta(\lambda_3) = -0.308 > -0.987 = \theta(\mu_3)$. **Eliminate left!**

$$[a_4, b_4] = [-1.832, 0.056], \text{ length} = 1.888.$$

Reuse: $\lambda_4 = \mu_3 = -1.112$, $\theta(\lambda_4) = -0.987$.

*Note: This is the first time we eliminate the **left** side instead of the right. The function value at λ_3 is worse because $\lambda_3 = -1.832$ is farther from the minimum $\lambda^* = -1$ than $\mu_3 = -1.112$.*



Iteration 4

Current interval: $[a_4, b_4] = [-1.832, 0.056]$, length = 1.888.

Known: $\lambda_4 = \mu_3 = -1.112$ with $\theta(\lambda_4) = -0.987$ (reused).

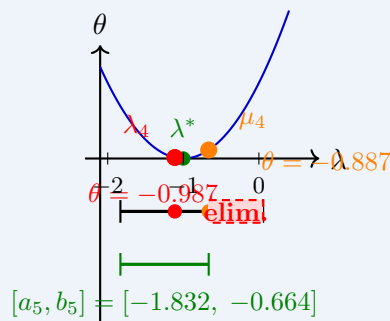
Compute new point:

$$\mu_4 = 0.382(-1.832) + 0.618(0.056) = -\mathbf{0.664}$$

$$\theta(\mu_4) = (-0.664)^2 + 2(-0.664) = 0.441 - 1.328 = -\mathbf{0.887}$$

Compare: $\theta(\lambda_4) = -0.987 \leq -0.887 = \theta(\mu_4)$. Eliminate right.

$$[a_5, b_5] = [-1.832, -0.664], \text{ length} = 1.168.$$

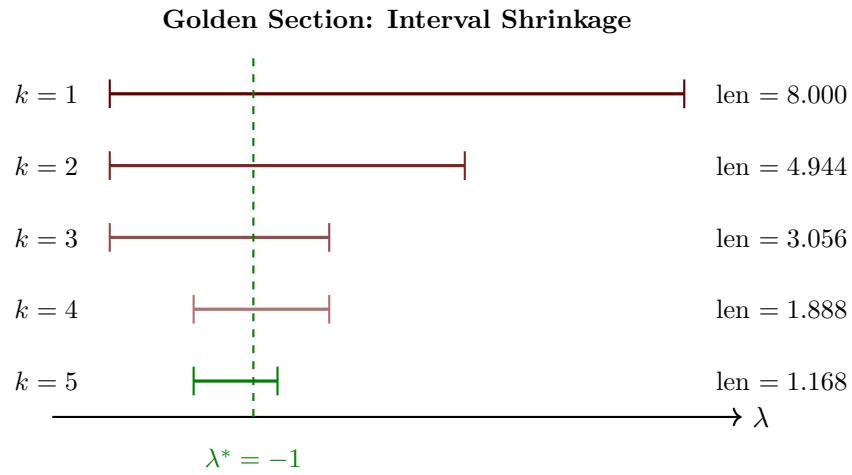
**Golden Section Summary After 4 Iterations**

Starting from $[-3, 5]$ (length 8), after 4 iterations and **5 total function evaluations**:

Iter.	$[a_k, b_k]$	Length	New evals	Action
1	$[-3, 5]$	8.000	2	Eliminate right
2	$[-3, 1.944]$	4.944	1	Eliminate right
3	$[-3, 0.056]$	3.056	1	Eliminate left
4	$[-1.832, 0.056]$	1.888	1	Eliminate right
5	$[-1.832, -0.664]$	1.168	—	Final

The interval shrank by a factor of 0.618 each step: $8 \rightarrow 4.94 \rightarrow 3.06 \rightarrow 1.89 \rightarrow 1.17$.

The true minimum $\lambda^* = -1$ lies inside every interval.



Exercise 1: Continue the Golden Section Search

Continue the golden section method on $\theta(\lambda) = \lambda^2 + 2\lambda$ starting from the interval $[a_5, b_5] = [-1.832, -0.664]$.

Iteration 5: $[a_5, b_5] = [-1.832, -0.664]$, length = 1.168.

Known: $\mu_5 = \lambda_4 =$ _____, $\theta(\mu_5) =$ _____ (reused).

Compute:

$$\lambda_5 = 0.618(\text{_____}) + 0.382(\text{_____}) = \text{_____}$$

$$\theta(\lambda_5) = (\text{_____})^2 + 2(\text{_____}) = \text{_____}$$

Compare: $\theta(\lambda_5)$ _____ $\theta(\mu_5)$. Eliminate _____ side.

$$[a_6, b_6] = [\text{_____}, \text{_____}], \text{ length} = \text{_____}$$

Iteration 6: $[a_6, b_6] = [\text{_____}, \text{_____}]$.

Known (reused): _____

Compute new point:

$$\text{_____} = 0.618(\text{_____}) + 0.382(\text{_____}) = \text{_____}$$

$$\theta(\text{_____}) = \text{_____}$$

Compare and eliminate: $[a_7, b_7] = [\text{_____}, \text{_____}]$, length = _____.

Hint: After iteration 6, the interval should have length ≈ 0.446 .

2 Fibonacci Search

Key Idea

The Fibonacci method is identical in structure to golden section, but uses **varying** ratios $\alpha_k = F_{n-k}/F_{n-k+1}$ instead of a constant $\alpha = 0.618$. Here n is the total budget of function evaluations (fixed in advance). This makes Fibonacci search **optimal** among all methods that evaluate θ at one new point per iteration.

Setup. We choose $n = 10$ function evaluations. The Fibonacci numbers we need are:

$$\begin{aligned} F_0 = F_1 = 1, \quad F_2 = 2, \quad F_3 = 3, \quad F_4 = 5, \quad F_5 = 8, \\ F_6 = 13, \quad F_7 = 21, \quad F_8 = 34, \quad F_9 = 55, \quad F_{10} = 89. \end{aligned}$$

Iteration 1

Current interval: $[a_1, b_1] = [-3, 5]$, length = 8.

Ratio: $\alpha_1 = F_9/F_{10} = 55/89 \approx \mathbf{0.6180}$.

Place test points (same formulas as Golden Section):

$$\lambda_1 = \alpha_1 a_1 + (1 - \alpha_1) b_1 = 0.6180(-3) + 0.3820(5) = \mathbf{0.0545}$$

$$\mu_1 = (1 - \alpha_1) a_1 + \alpha_1 b_1 = 0.3820(-3) + 0.6180(5) = \mathbf{1.9455}$$

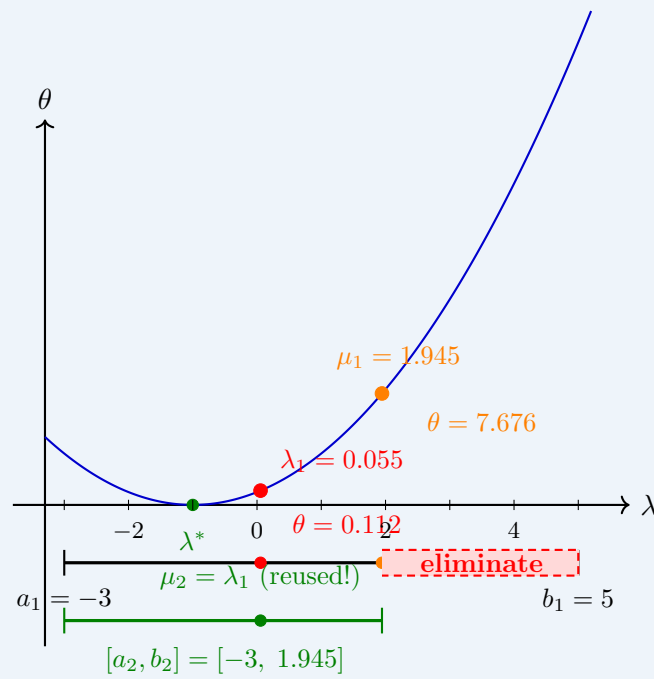
Evaluate: $\theta(\lambda_1) = \mathbf{0.112}$, $\theta(\mu_1) = \mathbf{7.676}$.

Compare & eliminate: $\theta(\lambda_1) \leq \theta(\mu_1)$, so discard the right:

$$[a_2, b_2] = [-3, 1.9455], \quad \text{length} = 4.9455.$$

Reuse $\mu_2 = \lambda_1 = 0.0545$ with $\theta(\mu_2) = 0.112$.

Note: $\alpha_1 = 0.6180$ is almost identical to the golden ratio, so this first step is nearly the same as Golden Section. The differences appear in later iterations where α_k varies.



Iteration 2

Ratio: $\alpha_2 = F_8/F_9 = 34/55 \approx \mathbf{0.6182}$. (differs from $\alpha_1 = 0.6180!$)

Known: $\mu_2 = \lambda_1 = 0.055$, $\theta(\mu_2) = 0.112$ (reused).

Compute:

$$\lambda_2 = 0.6182(-3) + 0.3818(1.945) = \mathbf{-1.109}$$

$$\theta(\lambda_2) = (-1.109)^2 + 2(-1.109) = 1.230 - 2.218 = \mathbf{-0.988}$$

Compare: $\theta(\lambda_2) = -0.988 \leq 0.112 = \theta(\mu_2)$. Eliminate right.

$$[a_3, b_3] = [-3, 0.055], \quad \text{length} = 3.055.$$

Reuse: $\mu_3 = \lambda_2 = -1.109$, $\theta(\mu_3) = -0.988$.

Iteration 3

Ratio: $\alpha_3 = F_7/F_8 = 21/34 \approx \mathbf{0.6176}$.

Known: $\mu_3 = -1.109$, $\theta(\mu_3) = -0.988$ (reused).

Compute:

$$\lambda_3 = 0.6176(-3) + 0.3824(0.055) = \mathbf{-1.836}$$

$$\theta(\lambda_3) = (-1.836)^2 + 2(-1.836) = 3.371 - 3.672 = \mathbf{-0.300}$$

Compare: $\theta(\lambda_3) = -0.300 > -0.988 = \theta(\mu_3)$. **Eliminate left!**

$$[a_4, b_4] = [-1.836, 0.055], \quad \text{length} = 1.891.$$

Reuse: $\lambda_4 = \mu_3 = -1.109$, $\theta(\lambda_4) = -0.988$.

Iteration 4

Ratio: $\alpha_4 = F_6/F_7 = 13/21 \approx \mathbf{0.6190}$.

Known: $\lambda_4 = -1.109$, $\theta(\lambda_4) = -0.988$ (reused).

Compute:

$$\mu_4 = (1 - \alpha_4)(-1.836) + \alpha_4(0.055) = 0.3810(-1.836) + 0.6190(0.055) = \mathbf{-0.665}$$

$$\theta(\mu_4) = (-0.665)^2 + 2(-0.665) = 0.442 - 1.330 = \mathbf{-0.888}$$

Compare: $\theta(\lambda_4) = -0.988 \leq -0.888 = \theta(\mu_4)$. Eliminate right.

$$[a_5, b_5] = [-1.836, -0.665], \quad \text{length} = 1.171.$$

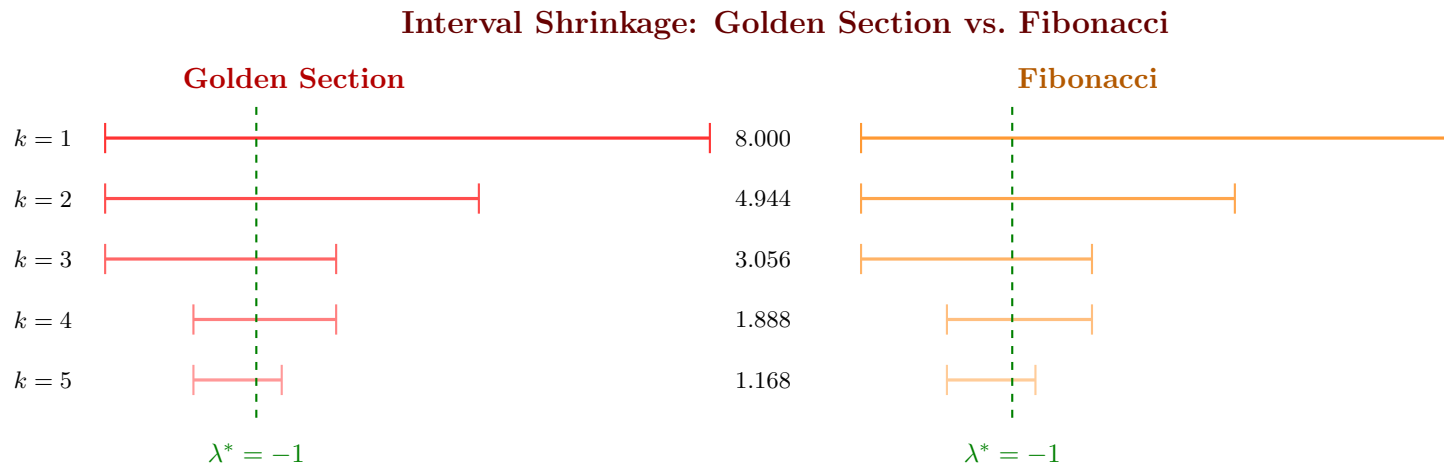
Fibonacci Summary After 4 Iterations

Starting from $[-3, 5]$ (length 8), after 4 iterations and **5 total function evaluations**:

Iter.	α_k	$[a_k, b_k]$	Length	Action
1	$55/89 \approx 0.6180$	$[-3, 5]$	8.000	Eliminate right
2	$34/55 \approx 0.6182$	$[-3, 1.945]$	4.945	Eliminate right
3	$21/34 \approx 0.6176$	$[-3, 0.055]$	3.055	Eliminate left
4	$13/21 \approx 0.6190$	$[-1.836, 0.055]$	1.891	Eliminate right
5	—	$[-1.836, -0.665]$	1.171	Final

Notice how α_k varies slightly from iteration to iteration (unlike the constant 0.618 in Golden Section).

3 Comparing the Two Methods



Key Takeaways

1. **Both methods converge at similar rates.** After 4 iterations (5 evaluations), Golden Section gives length 1.168 and Fibonacci gives length 1.171 — essentially identical at this stage.
2. **Constant vs. variable α :** Golden Section uses $\alpha = 0.618$ at every step. Fibonacci uses $\alpha_k = F_{n-k}/F_{n-k+1}$, which fluctuates slightly around 0.618.
3. **Fibonacci is optimal** for a fixed budget n : it achieves the smallest possible final interval among all sequential comparison-based methods. For large n , the difference from Golden Section is negligible.
4. **Both reuse one test point per iteration**, requiring only 1 new function evaluation per step (after the initial 2).

Exercise 2: Continue the Fibonacci Search

Continue the Fibonacci method on $\theta(\lambda) = \lambda^2 + 2\lambda$ from $[a_5, b_5] = [-1.836, -0.665]$ with budget $n = 10$.

Iteration 5:

$$\alpha_5 = F_5/F_6 = \underline{\hspace{1cm}} / \underline{\hspace{1cm}} \approx \underline{\hspace{1cm}}$$

Known (reused): $\underline{\hspace{2cm}}$

Compute new point:

$$\underline{\hspace{1cm}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\theta(\underline{\hspace{1cm}}) = \underline{\hspace{2cm}}$$

Compare and eliminate: $[a_6, b_6] = [\underline{\hspace{1cm}}, \underline{\hspace{1cm}}]$

Iteration 6:

$$\alpha_6 = F_4/F_5 = \underline{\hspace{1cm}} / \underline{\hspace{1cm}} \approx \underline{\hspace{1cm}}$$

Known (reused): $\underline{\hspace{2cm}}$

Compute new point and compare:

Exercise 3: Apply Golden Section to a Different Function

Consider the function $\theta(\lambda) = e^{-\lambda} + \lambda^2$ on the interval $[0, 2]$.

(This function has a minimum near $\lambda \approx 0.3517$.)

(a) Perform **Iteration 1** of the golden section method.

$$\alpha = 0.618. \quad [a_1, b_1] = [0, 2], \quad \text{length} = 2.$$

$$\lambda_1 = 0.618(0) + 0.382(2) = \underline{\hspace{2cm}}$$

$$\mu_1 = 0.382(0) + 0.618(2) = \underline{\hspace{2cm}}$$

$$\theta(\lambda_1) = e^{-\lambda_1} + \lambda_1^2 = \underline{\hspace{2cm}}$$

$$\theta(\mu_1) = e^{-\mu_1} + \mu_1^2 = \underline{\hspace{2cm}}$$

Compare: $\theta(\lambda_1) \underline{\hspace{1cm}} \theta(\mu_1)$. Eliminate $\underline{\hspace{1cm}}$ side.

$$[a_2, b_2] = [\underline{\hspace{1cm}}, \underline{\hspace{1cm}}]$$

(b) Continue for **Iteration 2** and **Iteration 3**.

(c) After 3 iterations (4 function evaluations), what is the length of your interval? How does the reduction ratio compare to $\alpha = 0.618$?